

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 2: Polynomials

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Zeroes and coefficients of a quadratic polynomial

1. α and β are the zeroes of $p(x) = x^2 - 3x - 1$. Find $\frac{1}{\alpha} + \frac{1}{\beta}$. **CBSE 2026, 2 marks**

SOLUTION

$$\alpha + \beta = 3 \text{ and } \alpha\beta = -1$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{3}{-1}$$

Answer: -3

2. Assertion (A): The polynomial $p(y) = y^2 + 4y + 3$ has two zeroes. Reason (R): A quadratic polynomial has at most two zeroes. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2026, 1 mark**

SOLUTION

$$y^2 + 4y + 3 = (y + 1)(y + 3), \text{ so the zeroes are } -1 \text{ and } -3. \text{ A is true.}$$

R is also true. But "at most two" only says there cannot be more than two zeroes. It does not show that this polynomial has exactly two.

Answer: (b)

3. The zeroes α and β of the polynomial $3x^2 + 6x + k$ satisfy $\alpha + \beta + \alpha\beta = -\frac{2}{3}$. Find k . **CBSE 2025, 1 mark**

SOLUTION

$$\alpha + \beta = -\frac{6}{3} = -2 \text{ and } \alpha\beta = \frac{k}{3}$$

$$-2 + \frac{k}{3} = -\frac{2}{3}, \text{ so } \frac{k}{3} = \frac{4}{3}$$

Answer: $k = 4$

4. Find the zeroes of the polynomial $x^2 + \frac{4}{3}x - \frac{4}{3}$. **CBSE 2025, 2 marks**

SOLUTION

$$x^2 + \frac{4}{3}x - \frac{4}{3} = \frac{1}{3}(3x^2 + 4x - 4)$$

$$3x^2 + 4x - 4 = 3x^2 + 6x - 2x - 4 = 3x(x + 2) - 2(x + 2) = (3x - 2)(x + 2)$$

The polynomial is zero when $3x - 2 = 0$ or $x + 2 = 0$.

Check: sum $= \frac{2}{3} - 2 = -\frac{4}{3}$ and product $= \frac{2}{3} \times (-2) = -\frac{4}{3}$, which match the coefficients.

Answer: $\frac{2}{3}$ and -2

5. The zeroes of $p(x) = 2x^2 - k\sqrt{2}x + 1$ add up to $\sqrt{2}$. Find k . (a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$
CBSE 2024, 1 mark

SOLUTION

$$\text{Sum of zeroes} = -\frac{b}{a} = \frac{k\sqrt{2}}{2}$$

$$\frac{k\sqrt{2}}{2} = \sqrt{2} \text{ gives } k = 2$$

Answer: (b)

6. Each zero of $x^2 + px + q$ is double a zero of $4x^2 - 5x - 6$. Find p . (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$ (c) -5 (d) 10
CBSE 2024, 1 mark

SOLUTION

Let the zeroes of $4x^2 - 5x - 6$ be α and β . Then $\alpha + \beta = \frac{5}{4}$.

The zeroes of $x^2 + px + q$ are 2α and 2β , with sum $2(\alpha + \beta) = \frac{5}{2}$.

Sum of zeroes $= -p$, so $p = -\frac{5}{2}$

Answer: (a)

2 Zeroes from the graph

7. The graph of $y = f(x)$ is a curve that stays above the x -axis everywhere and never touches it. How many zeroes does $f(x)$ have? (a) 0 (b) 1 (c) 2 (d) 4 **CBSE 2026, 1 mark**

SOLUTION

The zeroes are the x -coordinates of the points where the graph meets the x -axis.

This graph never meets the x -axis.

Answer: (a)

8. The graphs of two quadratic polynomials are drawn on the same axes. One parabola opens upwards and the other opens downwards, and both cut the x -axis at the same two points. How many distinct zeroes do the two polynomials have between them? **CBSE 2025, 1 mark**

SOLUTION

The zeroes of a polynomial are the x -coordinates of the points where its graph meets the x -axis.

Both graphs meet the x -axis at the same two points, so they share the same two zeroes.

Answer: 2

9. Assertion (A): a polynomial whose graph touches the x -axis at exactly one point cannot be quadratic. Reason (R): a polynomial of degree n , with $n > 1$, has at most n zeroes. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true. **CBSE 2024, 1 mark**

SOLUTION

A is false: the graph of $(x - 1)^2 = x^2 - 2x + 1$ touches the x -axis only at $x = 1$, and it is a quadratic polynomial.

R is true: a polynomial of degree n has at most n zeroes.

Answer: (d)

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.